Pre-design solution: Тепличный комплекс «Заря»

Region: Краснодарский край **Greenhouse type:** year-round **Crop:** tomato **Target yield:** 500.0 t/year **Generated:** 2026-06-17 06:34

1. Site inputs and analysis

Target yield per plot area: 25.0 kg/m²/year

Climate parameters (per Russian SP 131.13330):

Parameter	Value
Design winter temperature (t5)	-19.0 °C
Design summer temperature	29.4 °C
Heating degree-days	2510.0
Heating period	149 days
Snow load	1.0 kPa
Wind load	0.38 kPa
Solar radiation (winter)	3.8 MJ/m²/day

2. Design solution (variant v1)

Rationale: Year-round glass greenhouse complex for trellis tomato production in the Krasnodar region. Glass covering was chosen for its superior light transmittance ($\tau \approx 0.88$), which is critical for high-yield tomato cultivation during short winter days in the southern climate. A multi-span block layout is adopted to maximise the usable growing area within the 200 × 100 m plot while maintaining the structural rigidity required for year-round operation with full heating and supplemental lighting. Ridge height is set to 6.0 m to accommodate long trellis culture and overhead infrastructure (heating pipes, lighting bars, monorail). Soil type is loam with groundwater at 3.0 m depth, which is above the frost-heaving threshold, so a conventional strip foundation (depth 1.0 m) with standard waterproofing is recommended. The two main growing blocks are arranged along the plot length with a service road between them, and a full auxiliary zone (boiler house, storage, packaging, staff, admin, nutrient mixing) is grouped at one gable end.

Total growing area: 10368 m² **Complex footprint (with auxiliary):** 20000.0 m²

Greenhouse blocks

- **Block A** — 96.0 × 54.0 m (area 5184 m²)
- Heights: eave 4.0 m, ridge 6.0 m
- Layout: block (multi-span), 6 spans of 9.0 m
- Roof: straight, slope 45.0%
- Covering: glass (τ=0.88), glass 4.0 mm - Opaque structures share: 14.0%
- **Block B** — 96.0 × 54.0 m (area 5184 m²)
- Heights: eave 4.0 m, ridge 6.0 m
- Layout: block (multi-span), 6 spans of 9.0 m
- Roof: straight, slope 45.0%
- Covering: glass (τ=0.88), glass 4.0 mm - Opaque structures share: 14.0%

Complex structural parameters

Parameter	Value	SP Clause
Plinth height	0.4 m	5.8
Foundation rise above soil	0.4 m	5.9
Territory fence height	1.8 m	4.16

Auxiliary zones

- **Boiler house** — 120.0 m² (Gas-independent electric boiler plant and heat distribution for year-round heating)
- **Tool and fertiliser storage** — 80.0 m² (Storage of tools, consumables and mineral fertilisers)
- **Packaging and sorting hall** — 100.0 m² (Post-harvest handling, grading and packaging of tomato produce)
- **Staff facilities** — 60.0 m² (Locker rooms, showers, canteen and rest area for personnel)
- **Administrative building** — 80.0 m² (Offices, meeting room and security post)
- **Nutrient mixing and pumping room** — 60.0 m² (Preparation and distribution of nutrient solution for drip irrigation)

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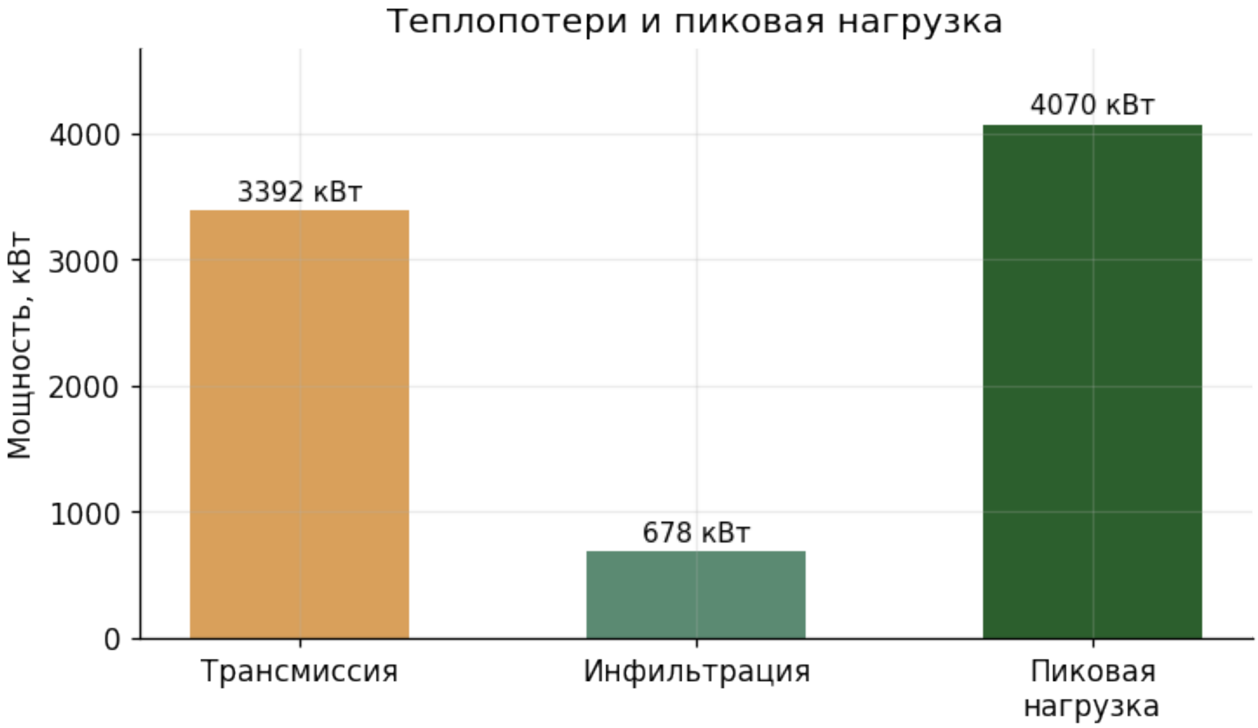
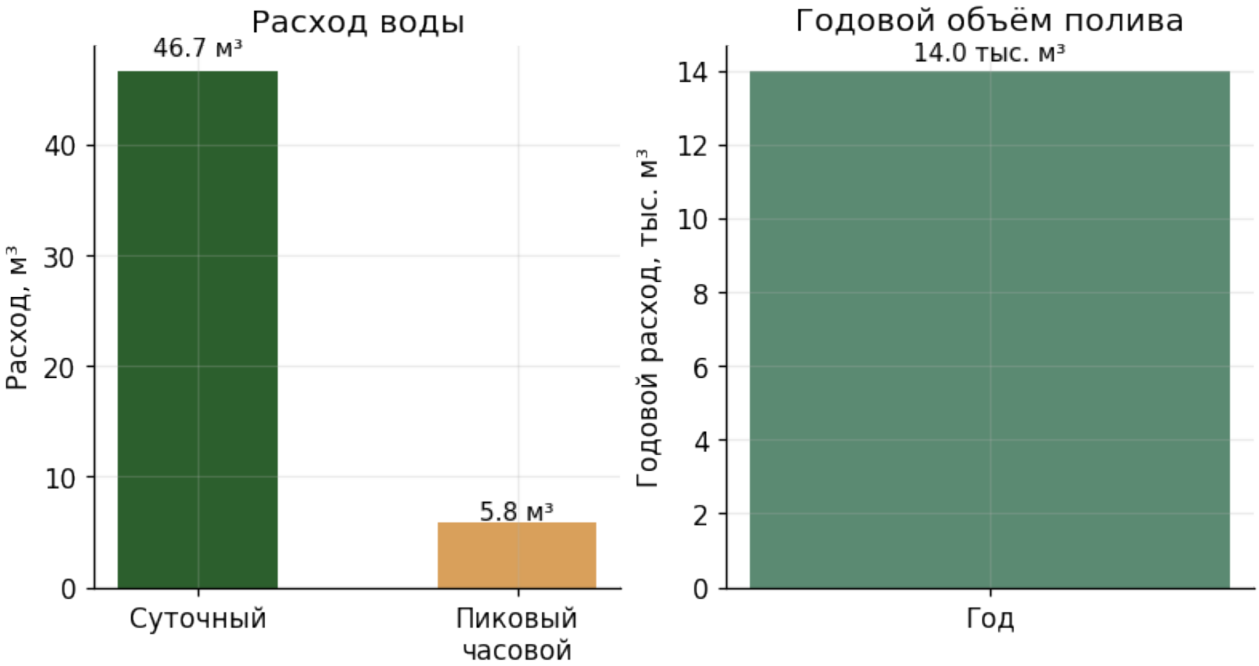
3. Engineering calculations

3.1. Heat supply



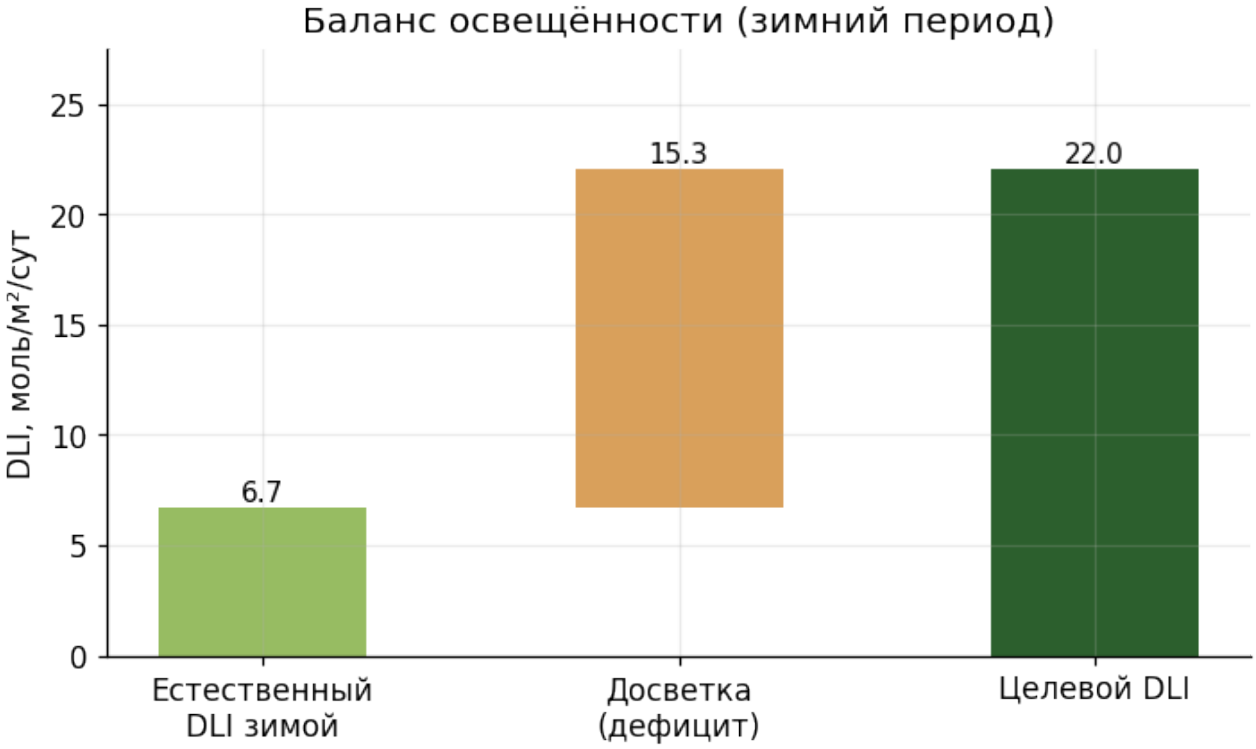
- Design temperature difference: **37.0 °C**
- Envelope area: **14323.2 m²**
- Overall heat transfer coefficient U: **6.4 W/(m²·K)**
- Transmission losses: **3391.7 kW**
- Infiltration losses: **678.3 kW**
- **Peak heating load: 4070.1 kW**
- Annual heat demand: **6626.5 MWh**
- Coolant temperature: **95.0 °C** (SP 7.9 ≤150)
- Lower-zone heat share: **45.0%** (SP 7.13 ≥40)

3.2. Water supply



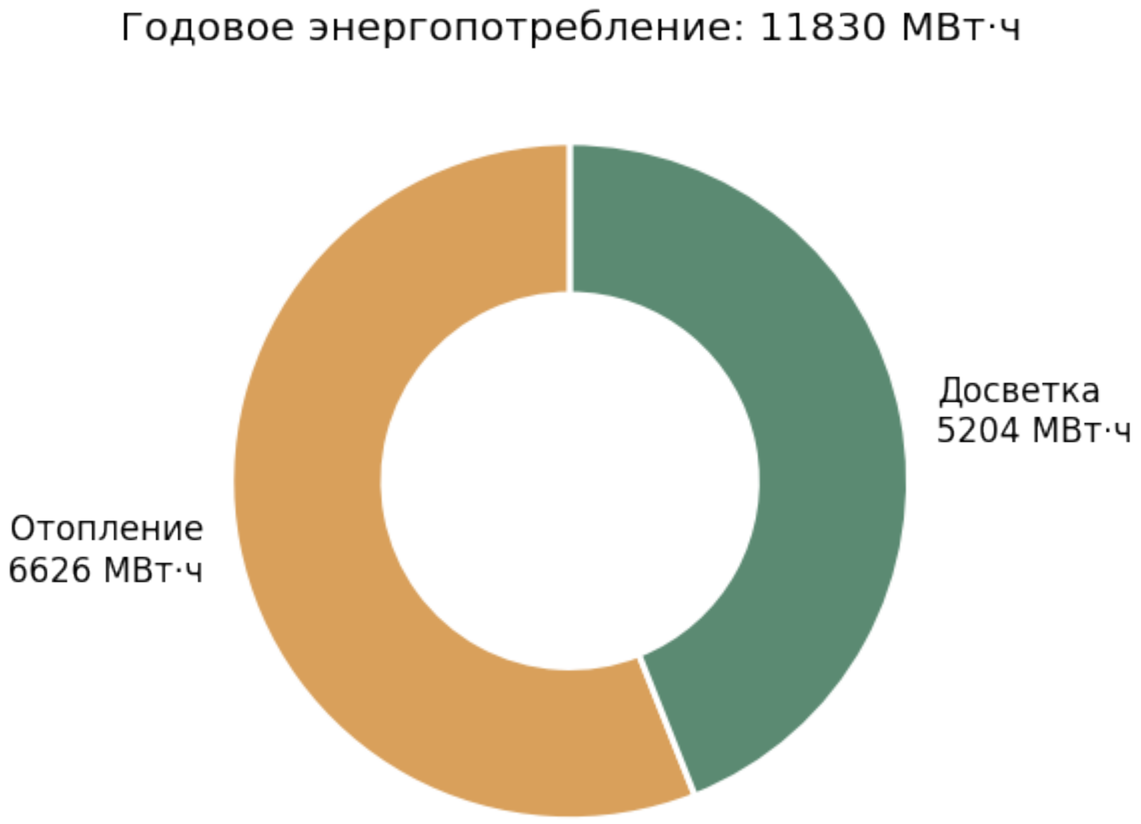
- Daily demand: **46.66 m³/day**
- Peak hourly: **5.83 m³/h**
- Annual: **13997 m³/year**
- Irrigation method: Drip (default)
- Reliability category: **II** (SP 6.14)
- Hose service radius: **40.0 m** (SP 6.8 ≤45)

3.3. Lighting and supplemental



- Target DLI: **22.0 mol/m²/day**
- Natural winter DLI: **6.7 mol/m²/day**
- **Supplemental lighting required**: installed 210.5 W/m², consumption 5203983 kWh/year
- Aisle floor illuminance: 8.0 lx (SP 8.3 ≤10)

3.4. Annual energy consumption



3.5. Ventilation

- Summer target air changes per hour: **60.0 h⁻¹**
- Vent opening share of envelope: **20.0%** (SP 7.18 ≥20, ≥10 north of 60°N)
- Vent opening share of floor: 27.6%
- Forced ventilation: **not required**

3.6. Structural loads

- Snow load on roof: **16692.0 kN** (γ=1.4)
- Wind load on walls: **730.0 kN** (q_{10}=1.0, q_2=0.6)
- Trellis load: 150.0 N/m² (γ=1.3)

Overload factors and normative values per SP 107.13330 clause 5.14.

3.7. Geotechnical conditions and foundation

- Foundation soil: **loam**
- Groundwater depth: **3.0 m**
- Recommended foundation type: **strip**
- Minimum embedment depth: **1.0 m**

- Perimeter drainage: **not required**
- Waterproofing grade: **standard**

> Loam — strip foundation with waterproofing; drainage if groundwater table is high.

Preliminary recommendations. A proper engineering-geological survey per SP 47.13330 is required for working documentation.

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4. SP 107.13330 compliance check

Rules checked: **34**

WARNING — ENG.2-aux-share

Подсобные зоны должны занимать 5–30% площади комплекса. Меньше — нехватка для обслуживания; больше — неэффективное использование участка.

- Actual: `2.5`
- Required: `[5.0, 30.0]`
- **Source** (SP 107.13330, original Russian text): Инженерная проверка (не из СП): обоснованная доля подсобок

Generated by agro-greenhouse-designer. Outputs are pre-design estimates and require validation by a qualified engineer before detailed documentation.